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Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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A la meva família, que ha estat al meu costat en tot moment, donant-me l'empenta i el suport incondicional que necessitava, sobretot quan més em costava creure en mi mateix. I als meus amics, per ser la força que m'ha acompanyat dia a dia. Aquest projecte no marca només el final del grau, sinó el tancament d'una etapa vital molt important que m'ha canviat com a persona i l'inici d'un nou capítol.

A l'Hospital de Sant Pau i al Dimension Lab, per obrir-me les portes i ensenyar-me que el més bonic d'aquesta professió no són els algoritmes ni els números, sinó veure com allò que aprens pot servir per ajudar les persones i fer el bé. Sentir que aquest esforç pot tenir un impacte real i útil en la vida dels altres és la recompensa que dona sentit a tot el camí.

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I further acknowledge the prior work of the students and researchers within this clinical ecosystem, upon which much of the technical framework of this thesis is built.

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Abstract

Coronary artery disease (CAD) assessment from coronary computed tomography angiography (CCTA) remains largely manual and time-consuming in clinical practice. At Hospital de la Santa Creu i Sant Pau, expert review of a single study can require between 30 minutes and two hours depending on case complexity. This bachelor's thesis presents an automated, end-to-end computational pipeline that transforms segmented coronary anatomy into quantitative stenosis metrics, CAD-RADS 2.0-oriented reporting support, and an interactive clinical visualization dashboard. The system is organized into four modular blocks: automated anatomy extraction, geometric stenosis quantification, segment labeling with CAD-RADS scoring, and a Streamlit-based decision-support interface. Designed as a transparent in-house framework rather than a proprietary black box, the pipeline aims to reduce reporting burden, improve reproducibility, and assist patient prioritization while preserving final clinical authority. Validation combines ASOCA and MACS-18 cohorts with synthetic phantoms for controlled geometric verification.

Resum

L'avaluació de la malaltia coronària (MAC) a partir de la tomografia computada coronària (TCCC) continua sent, en la pràctica clínica, un procés manual i molt costós en temps. A l'Hospital de la Santa Creu i Sant Pau, la revisió experta d'un estudi pot requerir entre 30 minuts i dues hores segons la complexitat del cas. Aquest treball de fi de grau presenta un pipeline computacional automatitzat d'extrem a extrem que transforma l'anatomia coronària segmentada en mètriques quantitatives d'estenosi, suport orientat al CAD-RADS 2.0 i un tauler clínic interactiu de visualització. El sistema s'organitza en quatre blocs modulares: extracció automatitzada de l'anatomia, quantificació geomètrica de l'estenosi, etiquetatge segmentari amb puntuació CAD-RADS i una interfície Streamlit de suport a la decisió. Dissenyat com a marc transparent i intern, el pipeline pretén reduir la càrrega de informes, millorar la reproduïbilitat i assistir la priorització de pacients sense substituir l'autoritat clínica final. La validació combina les cohorts ASOCA i MACS-18 amb fantasmes sintètics per a la verificació geomètrica controlada.

Resumen

La evaluación de la enfermedad coronaria (EC) a partir de la angiografía por tomografía computarizada coronaria (ATCC) sigue siendo, en la práctica clínica, un proceso manual y muy costoso en tiempo. En el Hospital de la Santa Creu i Sant Pau, la revisión experta de un estudio puede requerir entre 30 minutos y dos horas según la complejidad del caso. Este trabajo de fin de grado presenta un *pipeline* computacional automatizado de extremo a extremo que transforma la anatomía coronaria segmentada en métricas cuantitativas de estenosis, soporte orientado al CAD-RADS 2.0 y un panel clínico interactivo de visualización. El sistema se organiza en cuatro bloques modulares: extracción automatizada de la anatomía, cuantificación geométrica de la estenosis, etiquetado segmentario con puntuación CAD-RADS e interfaz Streamlit de apoyo a la decisión. Diseñado como un marco transparente e interno, el *pipeline* pretende reducir la carga de informes, mejorar la reproducibilidad y asistir la priorización de pacientes sin sustituir la autoridad clínica final. La validación combina las cohortes ASOCA y MACS-18 con fantomas sintéticos para la verificación geométrica controlada.

Preface

This document presents the bachelor's thesis completed during the 2025–2026 academic year within the Bachelor's Degree in Mathematical Engineering in Data Science at Universitat Pompeu Fabra, in collaboration with the Dimension Lab at Hospital de la Santa Creu i Sant Pau.

The main body of the thesis (Introduction through Conclusions) is written in English for consistency with the scientific literature in the field.

Use of Generative AI Disclosure

The Author, Adrià Cortés Cugat, discloses that the writing of this document "Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support" involved the use of Cursor AI and Google Gemini AI, for the purpose of: language review; text editing; technical writing support; LaTeX integration and formatting; software development assistance; code debugging and optimization; and pipeline documentation support. The Author warrants, however, that the contents of this thesis are a genuine human creation, as the results obtained from Cursor AI and Google Gemini AI have been duly reviewed and edited by Adrià Cortés Cugat. In accordance with university guidelines on the use of AI in assessable work, the technical and clinical substance of the main body, especially the Introduction, Methodology, and Conclusions, has been written, validated, and revised by the Author with the support of the thesis director and supervisor.

Information source disclosure. The Author declares that all cited sources have been appropriately reviewed and validated by Adrià Cortés Cugat to support the claims made in this work, ensuring they are sufficiently contrasted and reliable.

Note on Figures and Data

The diagram figures that illustrate clinical and reporting workflows have been entirely designed and created by the Author. Where other figures build upon internal hospital materials, they are explicitly cited in the corresponding figure captions. Clinical cohort data and pipeline results are documented in the corresponding chapters and, where appropriate, in the supporting appendix.

Contents

List of Figures

List of Tables

Chapter 1

INTRODUCTION

1.1. Clinical Context and Motivation

1.1.1. Anatomy of Coronary Arteries

The human heart is the central organ of the cardiovascular system and requires a continuous supply of oxygen and nutrients to function properly. This supply is provided by the coronary artery tree, a specialized vascular network that delivers blood directly to the heart muscle, known as the myocardium. The coronary arteries originate from the root of the aorta and surround the surface of the heart in a crown-like arrangement, from which the term *coronary artery* is derived [11]. As illustrated in Figure 1.1, the coronary arterial system is mainly composed of two vessels: the right coronary artery (RCA) and the left coronary artery (LCA).

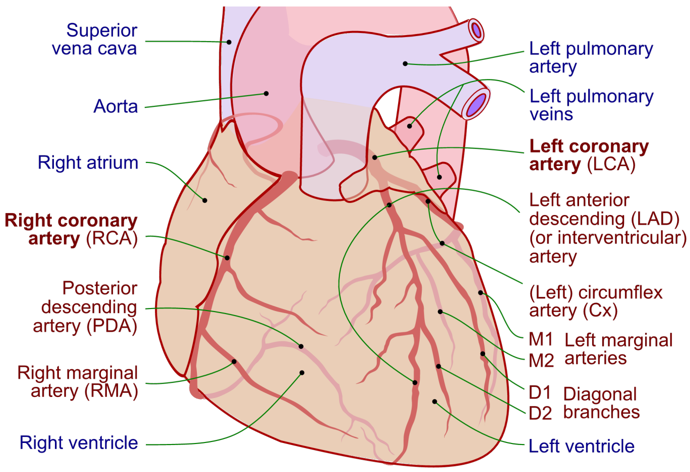

Figure 1.1: Anterior view of the human coronary tree. Coronary arteries are highlighted in red, and other cardiac structures in blue. Adapted from Lynch et al. [23].

The RCA primarily supplies blood to the right atrium and right ventricle. In contrast, the LCA quickly divides into two major branches: the left anterior descending (LAD) artery and the circumflex (LCx), which together supply the majority of the left ventricle and the interventricular septum [11]. From these primary vessels, a network of smaller branches extends deep into the myocardium to ensure regional blood delivery across the entire heart.

A critical characteristic of the coronary circulation is its limited collateral connectivity. Unlike other vascular systems in the human body, coronary arteries generally lack sufficient alternative pathways to redirect blood flow when an obstruction occurs. Consequently, any narrowing or blockage in a proximal arterial segment directly compromises blood flow to all downstream regions supplied by that vessel [26].

Additionally, coronary anatomy presents significant inter-patient variability in terms of vessel size, branching structure, and coronary dominance. Dominance is defined by which artery gives rise to the posterior descending artery (PDA) and can be classified as right-dominant, left-dominant, or codominant. Right-dominant circulation, in which the PDA originates from the RCA, is the most common configuration, followed by left dominance and codominance [29]. Understanding this complex anatomical layout is fundamental, as it provides the basis for locating lesions, quantifying stenosis severity, and stratifying risk in coronary artery disease (CAD) diagnosis.

1.1.2. Coronary Artery Disease

Coronary artery disease (CAD) remains one of the leading causes of morbidity and mortality worldwide [28]. It is characterized by the progressive accumulation of atherosclerotic plaque within the walls of the coronary arteries, leading to a narrowing of the vessel lumen known as stenosis. This narrowing restricts blood flow to the myocardium and, if left untreated, can result in myocardial ischemia or infarction [21, 24]. Figure 1.2 illustrates this pathological mechanism, contrasting a normal artery with one obstructed by plaque buildup.

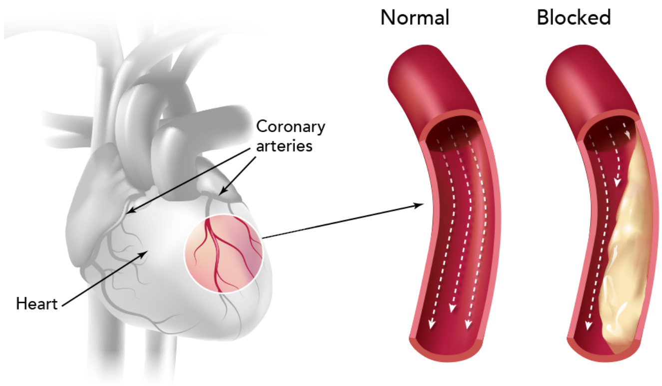

Figure 1.2: Coronary arteries narrowed by atherosclerotic plaque. Adapted from Kaiser Permanente [18].

Early and accurate diagnosis is essential for effective risk stratification and patient management. In current clinical practice, coronary computed tomography angiography (CCTA) is the standard non-invasive imaging technique for CAD evaluation [21]. CCTA provides detailed 3D reconstructions of the coronary tree, allowing clinicians to detect plaques, analyze their spatial location, and estimate the degree of luminal narrowing.

However, despite the high resolution of CCTA, the diagnostic workflow remains highly time-consuming and heavily dependent on manual and visual assessment. For instance, at Hospital de la Santa Creu i Sant Pau, reporting non-pathological or low-complexity cases requires approximately 30 minutes, while complex evaluations can take up to 2 hours [1]. This significant clinical burden drives the need for automated, data-driven tools capable of quantifying stenosis efficiently and supporting clinical decision-making, while fully preserving expert interpretability and clinical control.

1.1.3. Stenosis Quantification: A Key Indicator of CAD Severity

Among the various indicators used to assess coronary artery disease, stenosis severity plays a central role in clinical diagnosis and risk stratification. Stenosis represents the reduction of the vessel lumen caused by atherosclerotic plaque accumulation, a condition directly associated with an increased risk of myocardial ischemia and adverse cardiac events [14, 21]. In clinical practice, this severity is quantified using geometric measurements such as the minimal lumen area (MLA) or percentage diameter stenosis (%DS). These metrics evaluate the degree of narrowing by comparing the most constricted point of a lesion with a reference baseline, which is typically estimated from adjacent, relatively healthy vessel segments [14, 16].

Although these definitions are conceptually simple, accurate stenosis quantification presents significant clinical challenges. Coronary arteries exhibit complex, tortuous geometries, and diffuse atherosclerosis often makes it difficult to identify truly healthy reference regions. In addition, limitations in image resolution, segmentation errors, and noise in CCTA-derived data can highly influence measurements, leading to variability in stenosis estimates [16, 19].

In routine clinical practice, stenosis assessment is often performed visually or using semi-automatic tools provided by proprietary software [1, 12]. Although these methods are widely used, they depend heavily on the clinician's interpretation and can vary between observers, which reduces consistency and reproducibility [16].

Consequently, there is a strong clinical need to develop robust, transparent, and reproducible quantification methods. Automating these geometric measurements has the potential to standardize stenosis evaluation, reduce reporting time, and provide reliable quantitative data essential for downstream clinical tasks, such as CAD-RADS 2.0 assignment and patient prioritization.

1.1.4. CAD-RADS 2.0 and the Need for Decision Support

To standardize the reporting of coronary disease and facilitate clinical communication, the Coronary Artery Disease-Reporting and Data System (CAD-RADS 2.0) was established [12]. This structured framework categorizes patients into distinct risk scores ranging from 0 to 5, primarily based on the most severe stenosis measured via CCTA. Each categorical score reflects the overall disease burden and provides specific clinical recommendations, offering a standardized pathway for subsequent patient management [12].

While CAD-RADS 2.0 significantly improves reporting consistency across institutions, its assignment still relies heavily on the manual interpretation of stenosis severity across multiple vessels and segments. This process requires clinicians to continuously synthesize quantitative geometric measurements with complex anatomical context. Consequently, evaluating these scans becomes highly cognitively demanding, particularly in time-constrained clinical environments [1]. In practice, manual scoring remains susceptible to inter-observer variability, especially in borderline cases or in patients presenting with multiple moderate lesions. Such discrepancies can lead to inconsistent diagnostic classifications, ultimately impacting downstream clinical decisions, further testing, and patient prioritization [1, 16].

To address these limitations, there is a growing clinical imperative to develop automated decision support tools. These solutions can span from rule-based algorithms mapping quantitative stenosis measurements to machine learning approaches that leverage structured imaging features. Crucially, these computational tools aim to provide objective, reproducible, and transparent suggestions to assist radiologists and cardiologists. By streamlining the CAD-RADS assignment process, these systems reduce reporting times and variability while strictly preserving expert clinical judgment and final decision authority.

1.2. Clinical Automation Challenges at Hospital de la Santa Creu i Sant Pau

1.2.1. Current CAD Diagnostic Workflow and Associated Limitations

The current diagnostic workflow for Coronary Artery Disease (CAD) at Hospital de la Santa Creu i Sant Pau involves a complex and coordinated series of steps, managed primarily by the specialized cardiac imaging unit (Unitat d'Imatge i Funció Cardíaca, UIFC). As illustrated in Figure 1.3, the clinical pathway begins when a patient presenting with symptoms such as chest pain is scheduled for a Coronary Computed Tomography Angiography (CCTA). Once the scan is acquired, the images are stored in the hospital's *Picture Archiving and Communication System* (PACS) for further evaluation.

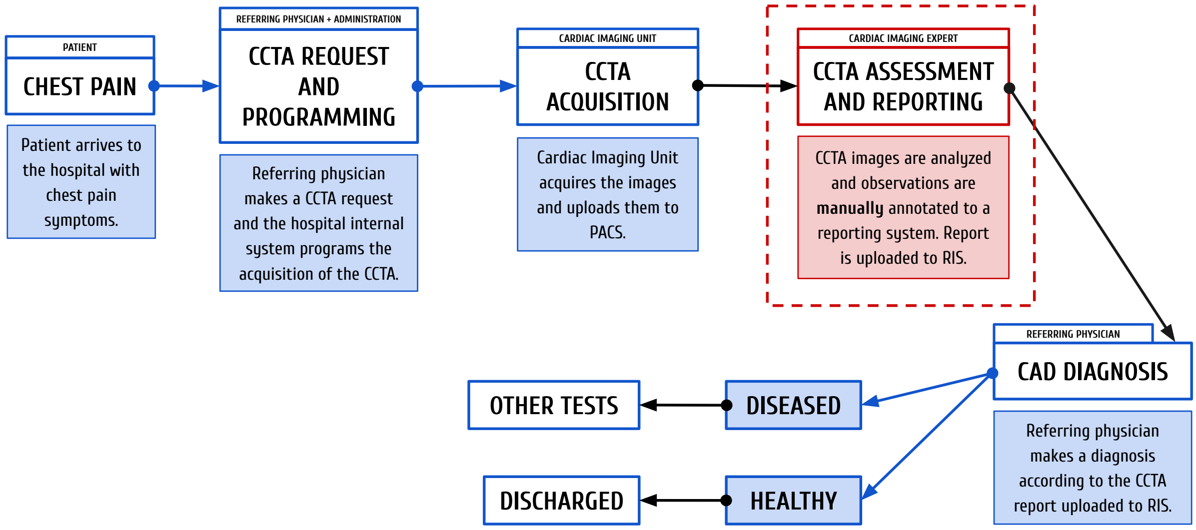

Figure 1.3: Current CAD diagnostic workflow at Hospital de Sant Pau. The primary operational bottleneck (image assessment and reporting) is highlighted in red. Adapted from Ferrer Beltran [13].

The core of this diagnostic pipeline relies heavily on the visual and manual analysis of these scans by expert cardiologists and radiologists to extract clinical metrics. These findings are then compiled into a clinical report, which the referring clinician evaluates to determine the appropriate subsequent medical actions, ranging from discharging a healthy patient to programming further invasive tests.

While the overall infrastructure is well-established, the central phase of image analysis and reporting (highlighted in red in Figure 1.3) represents a critical operational bottleneck.

1.2.2. The Bottleneck: Manual Image Analysis and Reporting

Zooming into the specific image analysis and reporting phase (Figure 1.3), the limitations of the current clinical pathway become evident. Currently, clinicians rely on commercial software solutions, such as *syngo.via*, to manually inspect the coronary arteries, differentiate segments, and calculate stenosis severity. Although these platforms offer semi-automated tools, the process remains highly dependent on the clinician's expertise to visually verify the vessels, identify bifurcations, and assess the plaque burden. Depending on the anatomical complexity and image quality, analyzing a single patient can consume around 30 minutes for healthy cases and 90 minutes for complex cases [1].

Once the visual assessment is complete, the extracted metrics are manually entered into an internal custom-built platform called *Filemaker*, which generates a template-based preliminary report. The specialist must then review, correct, and finalize this text before it is integrated into the *Radiology Information System* (RIS).

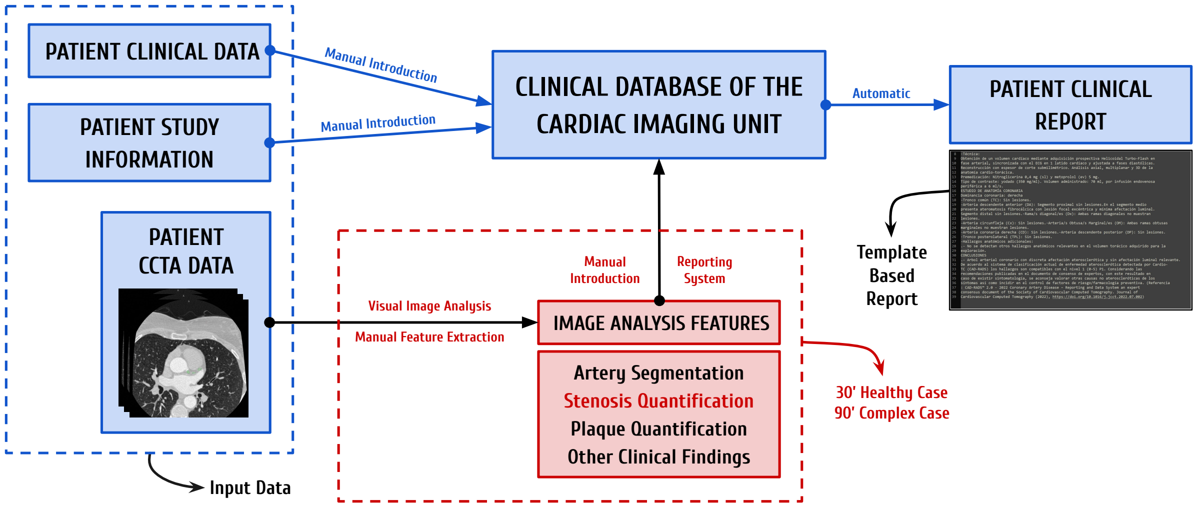

Figure 1.4: Detailed representation of the image analysis and reporting workflow at Hospital de Sant Pau. Adapted from Acebes Pinilla [1].

This heavy reliance on manual quantification and data entry presents significant operational challenges. Consequently, highly specialized experts spend considerable time performing repetitive, manual tasks on low-risk patients, rather than focusing their expertise on critical cases. This lack of an automated, prioritized workflow delays the diagnosis of severe cases and highlights a clear necessity for robust and automated solutions.

1.3. State of the Art: Automation of CAD Diagnosis and Visualization

The integration of Artificial Intelligence (AI) and advanced computer vision in cardiac imaging has seen exponential growth over the past decade, driven by the need to optimize clinical workflows [22]. Deep learning algorithms, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in medical image analysis, encouraging a move from manual CCTA assessment to automated systems [27]. However, significant limitations remain regarding their practical implementation. For these new techniques to succeed, they must be practical for hospitals, integrate smoothly into current systems, and be easily adopted by the clinical staff [20].

The prerequisite first step in any automated CAD assessment pipeline is the accurate segmentation and anatomical labeling of the coronary artery tree; before any downstream diagnostic metric can be calculated, the vessels must be correctly isolated and identified. Recent advancements have leveraged deep learning architectures, such as 3D U-Nets, to extract the vessel lumen from CCTA images [8, 31], while graph-based models like Graph Convolutional Networks (GCNs) are increasingly used to ensure topological consistency when labeling segments according to standard clinical guidelines [30]. Within Hospital de la Santa Creu i Sant Pau, preliminary research has explored these initial segmentation and labeling tasks using transformer-based models and the nn-UNet framework [9, 25].

Following successful segmentation, extracting the precise geometric properties of the vessels is essential. The Vascular Modeling Toolkit (VMTK) stands out as one of the most prominent open-source libraries for medical geometry data extraction [3]. Specifically, algorithms implemented within VMTK are widely utilized to extract accurate vascular centerlines and compute cross-sectional areas. These metrics provide the essential topological framework required to perform continuous Multi-Planar Reformations and subsequently evaluate vessel narrowing.

Once the 3D geometry of the coronary tree is established, the focus shifts to quantifying the disease burden. Current stenosis quantification techniques range from traditional intensity-based thresholding (which often struggles with severe calcifications) to modern machine learning models that predict stenosis severity directly from image features [17, 31]. Alternatively, geometric, non-ML approaches offer highly interpretable and robust methods by calculating the percentage of diameter or area stenosis based purely on the physical vessel profile, comparing the minimal lumen area against an interpolated healthy reference [16]. Recognizing the clinical value of algorithmic transparency, ongoing research at Hospital de la Santa Creu i Sant Pau has actively explored these purely geometric techniques to quantify stenosis, establishing a reliable, explainable foundation for disease assessment [7].

Beyond quantification, the effective visualization of these complex 3D metrics remains a critical challenge in the state of the art [6]. In the current clinical routine at Hospital de Sant Pau, specialists must manually navigate through cross-sectional slices and use standard viewing software to visually synthesize the data, assess the lesions, and manually fill out structured reports [1]. This cognitive load highlights a significant gap: the lack of dedicated, interactive visualization tools that bridge the gap between automated metrics and clinical reality. Modern research and clinical consensus emphasize that integrating 3D anatomical models alongside continuous luminal profiles and real medical images (such as Curved Multi-Planar Reformations) into a unified graphical interface is essential. Such integration allows clinicians to continuously validate automated findings against the ground-truth CCTA, drastically reducing the time and effort required to verify stenosis locations and seamlessly generate diagnostic reports [6].

Several commercial software solutions have emerged to address these overarching needs, including platforms like *Cleerly* [10] and *Artrya Salix* [4], which use AI to analyze arterial plaque and assess ischemia risk from CCTA images. While these robust dashboards offer highly desirable visualization components, they often function as proprietary "black boxes". This lack of algorithmic transparency significantly hinders explainability, a feature strongly demanded by medical experts to verify how a specific CAD-RADS score, plaque distribution, or stenosis percentage was computed. Finally, these standalone commercial platforms frequently struggle to integrate well into the highly specific IT infrastructures and patient prioritization workflows of public hospitals [20].

1.4. Project Scope and Contributions

To address the clinical challenges outlined in Section 1.2, and to overcome the limitations of commercial tools discussed in Section 1.3, this project contributes directly to the comprehensive, transparent, and inhouse AI-enabled diagnostic framework developed by the Dimension Lab at Hospital de la Santa Creu i Sant Pau [1]. While previous academic efforts have tackled earlier stages of this pipeline, such as patient prioritization [13] and coronary artery segmentation and labeling [9], a critical gap remains in translating raw segmented geometries into actionable clinical insights.

Therefore, the scope of this bachelor's thesis focuses on automating the geometric quantification of coronary stenosis, predicting the associated CAD-RADS score, and integrating these findings into an interactive clinical dashboard.

Figure 1.5 provides a high-level overview of where this project sits within the pipeline, while the detailed technical workflow is reserved for the Methodology chapter.

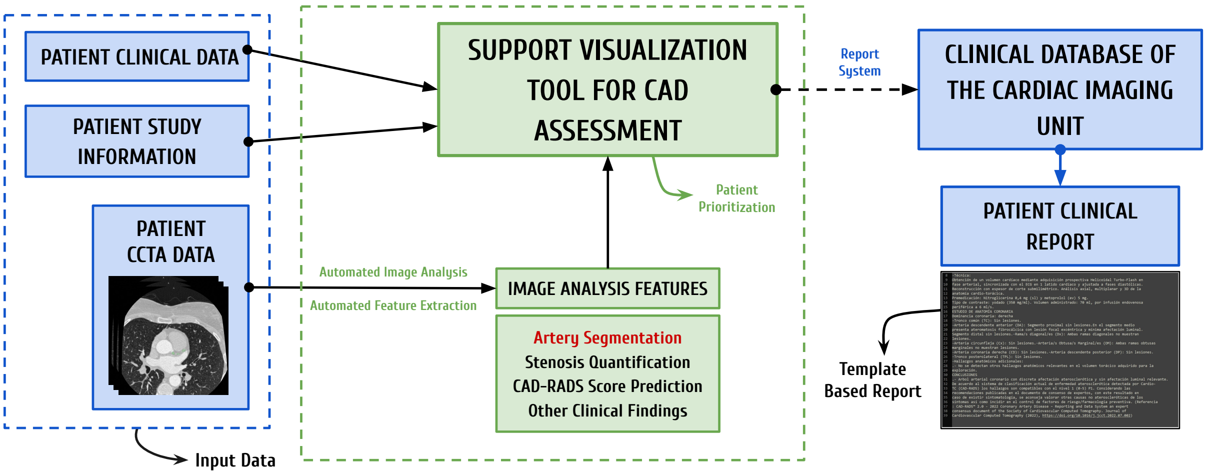

Figure 1.5: Contextual overview of the proposed automated diagnostic framework at Hospital de la Santa Creu i Sant Pau. The contributions of this thesis (green dashed box) bridge the gap between existing hospital data systems (blue) and the final clinical report. Adapted from Acebes Pinilla [1].

By seamlessly connecting raw image analysis with the hospital's reporting systems, this project aims to mitigate operational bottlenecks and reduce overall diagnostic times.

However, it is crucial to emphasize that the solution developed in this project is intended solely as a clinical decision-support tool, not as a replacement for medical expertise. In the highly sensitive healthcare sector, human oversight is crucial. Rather than replacing detailed diagnostic software, this visualization tool is designed to act as a rapid initial assessment interface. By presenting an immediate, high-level overview of the patient's critical metrics, it assists specialists in quickly evaluating prioritized patient lists, helping them decide which cases require immediate, in-depth analysis. Thus, the automated findings are designed to streamline the workflow, ensuring that the final diagnostic decisions and clinical validations always remain firmly in the hands of the medical professionals.

1.5. Objectives

The primary objective of this thesis is to design and develop an automated, end-to-end pipeline that successfully bridges the current gap in the CAD diagnostic workflow. Importantly, this project does not seek to maximize the precision of individual algorithmic components; rather, the proposed functional workflow is, in itself, the core product of this work.

To achieve this overarching goal, the project addresses a sequence of intermediate objectives. First, it requires deep contextualization within the complex clinical environment to evaluate prior research conducted at the hospital and design a unified, clinically applicable framework. Building upon this foundation, the project aims to implement computational geometry algorithms to accurately extract cross-sectional areas and calculate the percentage of area stenosis from 3D models. Subsequently, a standardized logic is developed to translate these geometric metrics into a reliable CAD-RADS score prediction.

To aggregate these findings and provide cardiologists with interpretable metrics, an interactive clinical dashboard is built. Functioning as a rapid-access pop-up interface, this Support Visualization Tool gives clinicians an immediate, visual summary of the patient's coronary status, ultimately reducing medical workload, optimizing case triaging, and significantly decreasing overall assessment time. Finally, a technical validation is conducted to verify the reliability of the extracted metrics and ensure the overall stability of the complete framework.

Chapter 2

METHODOLOGY

2.1. System Overview and Core Automated Workflow

This section formalizes the modular Python pipeline introduced in Section 1.4, translating the clinical decision-support framework into an executable computational workflow. For each patient identifier, a pre-segmented coronary lumen mask is processed sequentially into quantitative stenosis metrics, a CAD-RADS 2.0-oriented report, and an interactive visualization session.

The workflow comprises four automated blocks (Sections 2.3 to 2.6) executed as a linear chain, with each stage consuming the artifacts produced by its predecessor and no manual intervention between steps. A unified patient identifier routes ASOCA, MACS-18, and synthetic cohorts through the same implementations, while runtimes and extraction metrics are logged for Section 2.7.

2.1.1. End-to-End Data Flow

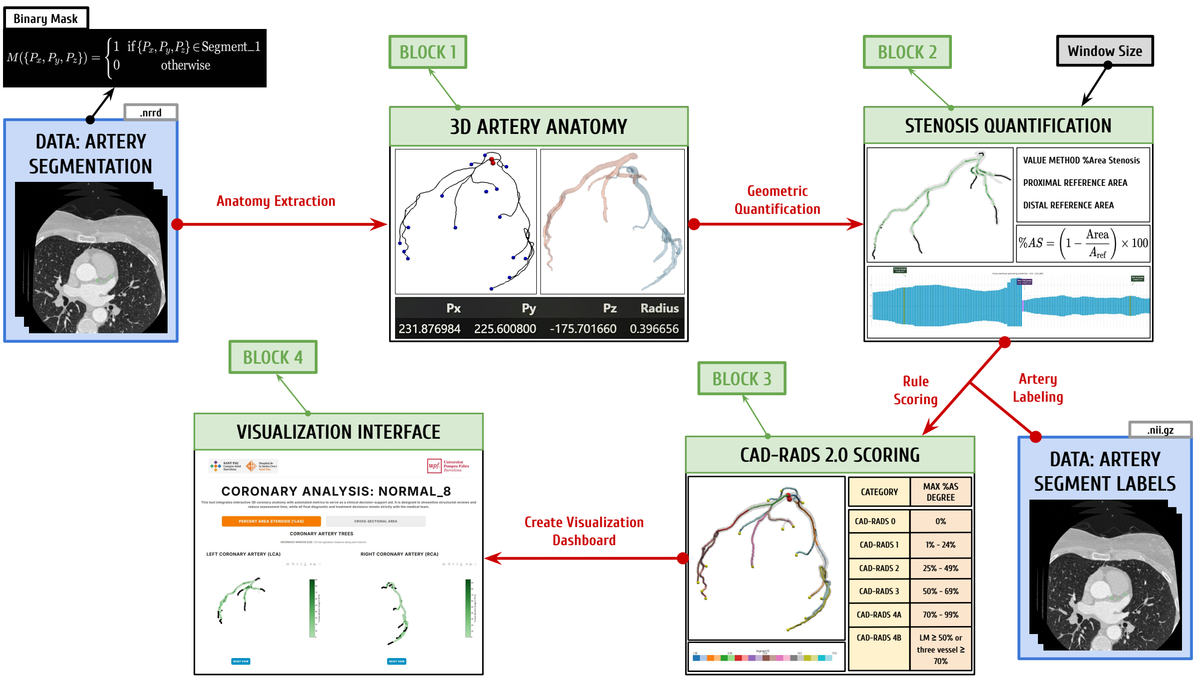

Figure 2.1: End-to-end automated workflow for a single patient case. Each block constitutes a self-contained processing stage; arrows indicate mandatory sequential dependencies and the principal data passed between stages.

Figure 2.1 summarizes the four-block architecture. The workflow assumes as input a pre-segmented coronary lumen mask and, when available, anatomical segment labels from prior segmentation and labeling work at the hospital [9].

Table 2.1: Overview of the four pipeline blocks: processing role, primary inputs, and principal outputs per patient case.

Block	**Primary input(s)**	**Principal output(s)**
1 Anatomy Extraction	Binary lumen mask	Centerline geometry, branch paths, and vessel surface models
2 Stenosis Quantification	Centerline and surface representations from Block 1	Cross-sectional area profiles, percentage area stenosis (%AS), and merged branch-level data
3 Labeling & CAD-RADS	Geometric and stenosis tables	Segment stenosis summary and patient CAD-RADS report
4 Visualization	Aggregated metrics and geometry from Blocks 1–3	Interactive clinical decision-support dashboard

To ensure explainability, the system retains intermediate mesh and tabular representations for potential clinical audits. The clinical cohorts utilized, along with the methodological depth of each processing block, are detailed below.

2.2. ASOCA and MACS-18: Coronary Segmentation and Labeling Data

In medical image analysis, *segmentation* denotes the partition of a three-dimensional scan into meaningful regions. For each voxel in the volume, an expert or algorithm decides whether it belongs to a structure of interest, such as the blood-filled interior (*lumen*) of the coronary arteries, rather than to background tissue, myocardium, or surrounding anatomy. The result is not a single measurement but a spatial map that localizes the vessel tree within the patient-specific CCTA grid. The ASOCA and MACS-18 resources provide this map, together with companion anatomical label volumes, for a shared cohort of 40 CCTA cases (20 normal and 20 diseased anatomy). These data served as the reference on which Blocks 1–4 were designed and implemented. Once the pipeline was operational, the full cohort was processed to assess end-to-end behavior. The experimental protocol and result analysis are reported in Section 2.7.

2.2.1. Segmentation and Label Volumes

For each patient case, two aligned three-dimensional grids are provided, both registered to the physical coordinate system of the source CCTA so that scale and orientation are preserved.

The *segmentation annotation* is stored as a voxel grid in NRRD format. Each voxel indicates whether it belongs to the coronary lumen or to background tissue. The pipeline treats this volume as a binary mask.

The *segment label volume* is a separate grid, stored as a compressed NIfTI file, aligned with the segmentation. Instead of a binary lumen flag, each foreground voxel stores an integer class identifier that maps to a standardized coronary segment under the AHA 17-segment model [5]. Background voxels are assigned zero.

2.2.2. Datasets: ASOCA and MACS-18

The ASOCA (*Automated Segmentation of Coronary Arteries*) challenge established one of the first large, standardized benchmarks for fully automatic segmentation of complete normal and diseased coronary trees from CCTA [15]. The resource addresses the difficulty of imaging small, moving vessels. ASOCA is publicly available and provided the original lumen masks for the 40-patient cohort. In this project, those masks served as the working reference during pipeline development and remain the default segmentation supplied to Block 1.

MACS-18 (*Multiclass Anatomical Coronary Segmentation*) is a re-segmentation of the same 40 ASOCA patient volumes, currently in development at Hospital de la Santa Creu i Sant Pau and supervised by César Acebes Pinilla [1, 2]. Unlike ASOCA, this resource is not yet publicly available. Widely used public datasets have helped train automatic segmentation methods, but their lumen masks are often evaluated mainly by how well voxels match a reference, not by whether the vessel tree forms a continuous, anatomically plausible path [2]. When the lumen mask contains breaks or fails to follow the full length of the branches, extracting reliable centerlines and quantifying vessel geometry in Blocks 1 and 2 becomes more difficult. Expert manual refinements in MACS-18 target continuous lumen connectivity from the aortic root to the most distal visible branches, yielding more anatomically faithful masks without altering the underlying imaging data, patient identity, or labeling scheme. In this project, the same AHA segment label volumes are used for both ASOCA and MACS-18 cases; only the lumen mask differs between the two sources. The pipeline implementation is otherwise identical for both datasets.

2.3. Block 1: 3D Artery Anatomy Extraction

Block 1 transforms the binary lumen mask into robust 3D vessel surfaces and mathematical centerlines. By extracting global RCA and LCA geometries alongside isolated paths from the ostium to each distal tip, it establishes the anatomical framework required for stenosis quantification in Block 2.

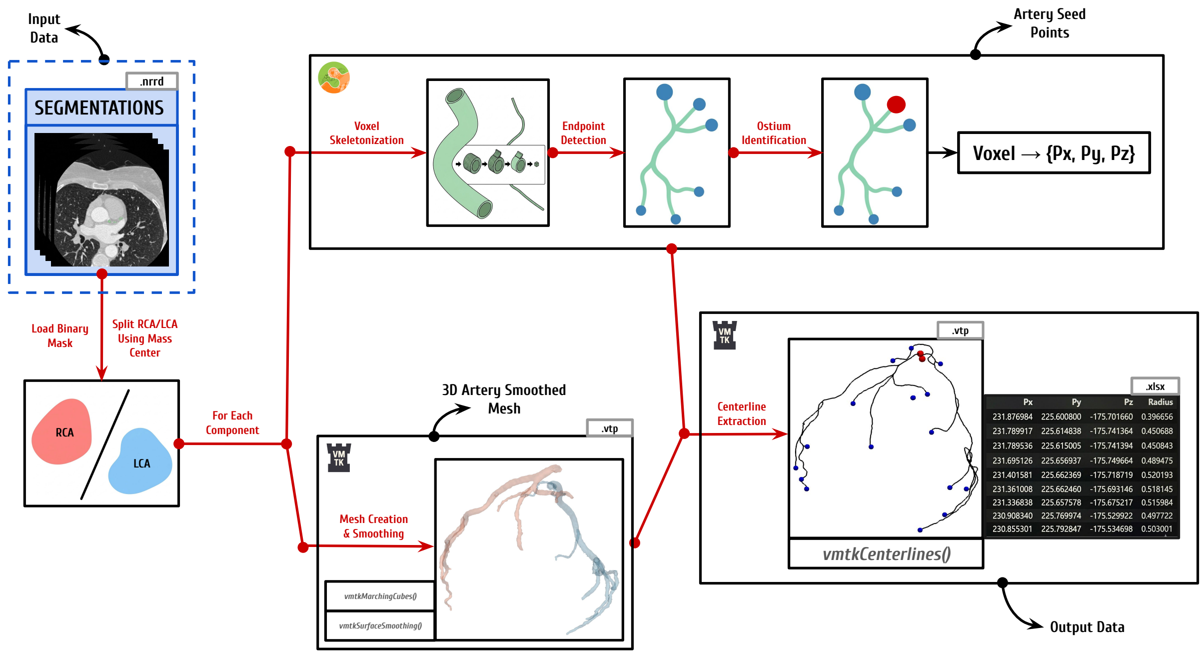

Figure 2.2: Block 1 hybrid workflow: spatial separation of the right and left coronary trees, automated seed placement on each tree, and extraction of centerline geometry with branch packaging.

As illustrated in Figure 2.2, this block implements a hybrid approach. Instead of relying on a single tool, it combines automated voxel analysis with the surface and centerline extraction algorithms of the Vascular Modeling Toolkit (VMTK). This design completely removes the need for manual seed-point selection while maintaining the precise 3D shape of the smoothed vessels.

Phase 1: Spatial RCA/LCA Separation

The process begins by reading the binary lumen mask using `vmtkImageReader` to preserve voxel spacing and physical origin. To process the Right Coronary Artery (RCA) and Left Coronary Artery (LCA) independently, `scipy.ndimage.label` is applied to isolate the two largest connected components. In order to correctly classify these isolated components into the main RCA and LCA trees, the algorithm calculates their spatial center of mass. This metric represents the average 3D coordinate of all voxels within each object. By sorting these center points along the patient's spatial x-axis, the system follows standard anatomical conventions: the rightmost component is classified as the RCA, and the leftmost as the LCA. This splitting step produces two independent binary sub-volumes. Processing them separately is logically necessary before geometric extraction, as VMTK algorithms require a single starting source per connected surface.

Phase 2: Per-Artery Surface Reconstruction and Automated Seeding

The two independent binary sub-volumes outputted by Phase 1 serve as the direct input for the surface reconstruction. For each isolated tree, a 3D surface mesh is generated using `vmtkMarchingCubes`. This function creates a continuous geometric boundary formed by connected triangles by interpolating the exact boundary between the foreground lumen voxels and the background. The raw mesh is then refined via `vmtkSurfaceSmoothing` using a pass-band filter. In this context, a pass-band filter works by separating high-frequency noise from the actual anatomical structure. It removes the rough, blocky edges caused by the original voxels and this creates a clean, mathematically stable surface for the next calculations without altering the true size of the artery.

The next step requires placing seed points, which are the explicit starting and ending spatial coordinates required by VMTK to trace the centerlines. To fully automate this placement and eliminate manual user interaction, the system analyzes the overall 3D shape of the voxel mask to automatically find key structural landmarks. The algorithm to obtain these seed points works by iteratively "peeling" away the outer layers of the voxel grid (a process known as skeletonization) and was leveraged from prior work [9]. The volume is thinned using `skimage.morphology.skeletonize`, reducing the complex 3D shape down to a one-voxel-wide medial axis. Topological endpoints are then easily identified as voxels that have exactly one neighboring voxel (a node degree of one).

The ostium, where the artery leaves the aorta and centerline tracing must begin, is selected among the skeleton endpoints using the AHA label volume (Section 2.2.1). Under the AHA 17-segment model, segment 1 identifies the proximal RCA and segment 5 the proximal LCA [5]; an endpoint inside either region is designated as the ostium point. If none qualifies, the widest endpoint by maximum inscribed sphere radius is used as a fallback.

All remaining endpoints are designated as distal targets. Because these endpoints are initially identified in a discrete voxel grid (i, j, k), they must be mapped to the continuous physical space (x, y, z) to be compatible with the mesh. This is achieved using the image origin (O_x, O_y, O_z) and the voxel spacing (S_x, S_y, S_z) through the following spatial transformation:

$$x = O_x + i\,S_x, \quad y = O_y + j\,S_y, \quad z = O_z + k\,S_z. \tag{2.1}$$

Once mapped, these physical coordinates are projected onto the nearest vertex of the smoothed mesh to ensure the seed points lie exactly on the mathematical manifold.

Phase 3: Centerline Extraction, Topology Labeling, and Branch Packaging

With the seed points mapped onto the mathematical manifold, `vmtkCenterlines` is executed on the smoothed surface mesh. By providing the identified ostium as the source and the distal targets as the endpoints, the algorithm extracts the centerlines for each component of the coronary tree. These centerlines are 1D mathematical curves tracing exactly through the middle of the 3D vessels, creating a connected network that accurately maps the complex branching structure of the arteries. At every discrete point along these curves, the algorithm also calculates the `MaximumInscribedSphereRadius`, providing an initial estimation of the local vessel width.

Once the centerlines are generated, a labeling step assigns a specific topological point type to every coordinate based on its structural position. To guarantee anatomical validity for downstream stenosis quantification, the centerlines are partitioned into continuous branch paths running directly from the ostium to each distal endpoint. Geometrical artifacts are then removed by applying a strict filtering logic: paths containing fewer than 20 points, originating more than 4 mm from the true ostium, or failing to terminate at a valid endpoint are automatically discarded.

Finally, the extracted data is packaged into a structured format. The spatial coordinates (P_x, P_y, P_z), local radius, assigned point type, and corresponding AHA segment identifiers are stored as ordered tables for both the individual branches and the global coronary tree. The generated surface and centerline meshes are also saved in `.vtp` format for visual rendering. Concluding Block 1, the RCA and LCA tables are concatenated into a unified patient record, providing the exact geometric dataset required for Block 2.

2.4. Block 2: Geometric Stenosis Quantification

Block 2 transforms the Block 1 centerline and surface representations into percentage area stenosis (%AS) profiles by measuring cross-sectional lumen area and comparing each point against a locally interpolated healthy reference.

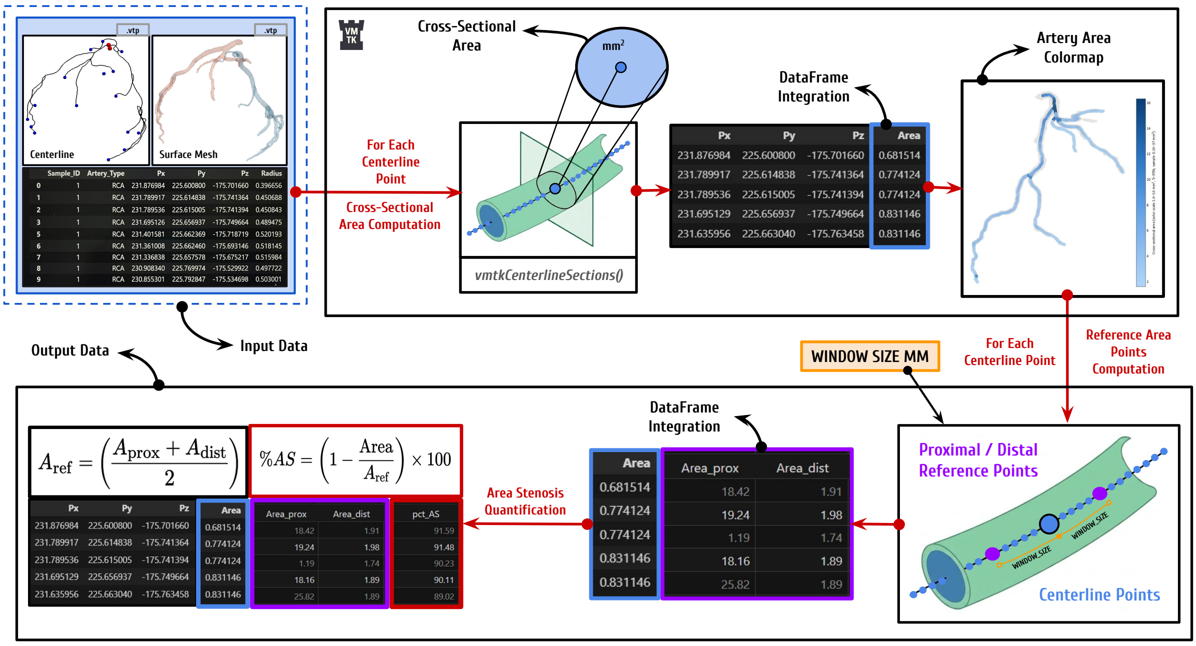

Figure 2.3: Block 2 workflow: orthogonal cross-sectional area extraction on each artery centerline and surface mesh, tabular integration, proximal/distal reference estimation, and %AS quantification.

As illustrated in Figure 2.3, Block 2 processes each available artery (RCA and LCA) independently. For every tree, it consumes the paired centerline and smoothed surface meshes produced in Section 2.3.

The computational logic for stenosis quantification and the methodology used to identify local healthy reference values build upon prior internal research by Burrull [7], originally grounded in the standard percentage diameter stenosis (%DS) framework described by Hideo-Kajita et al. [16]. Rather than reproducing diameter-based scores directly, this pipeline adapts those reference principles to cross-sectional *area* and percentage area stenosis (%AS).

In coronary arteries, atherosclerotic plaque frequently develops eccentrically rather than as a perfect concentric ring. Relying solely on one-dimensional diameter estimates, such as the Maximum Inscribed Sphere Radius (MISR), can therefore misrepresent luminal narrowing. In an elliptical or irregularly narrowed vessel, the inscribed sphere captures only the narrowest physical constraint and ignores the lateral space through which blood may still flow.

Conversely, the cross-sectional area captures the absolute reduction of the luminal opening regardless of cross-sectional shape. This yields a more robust representation of flow restriction and a geometric metric that correlates more closely with hemodynamic pressure drop than diameter alone [16]. The following phases mirror the diagram, detailing execution from localized area computation through patient-level aggregation.

Phase 1: Cross-Sectional Area Extraction

The process begins by lightly smoothing the global centerline and resampling it to establish a uniform spatial resolution. The polyline is interpolated so that the physical distance between consecutive discrete points is standardized to 0.1 mm by default. While larger intervals could reduce computational load, this spacing is deliberately chosen to capture high-fidelity geometric detail and subtle anatomical variations along the vessel. For exceptionally long arterial trees, this spacing is adaptively increased to ensure the total number of evaluated points does not exceed a maximum limit of 12,000 per component.

At every resampled centerline point, `vmtkCenterlineSections` computes the cross-sectional lumen area at that specific point. To do this, the algorithm finds the local direction (tangent) of the centerline and creates a flat, perpendicular cutting plane. It then slices the 3D smoothed surface mesh (from Block 1) with this plane to extract the exact 2D boundary of the vessel lumen, calculating its enclosed area in mm^2. Any incomplete or open slices are automatically discarded.

Highly tortuous or irregular geometries can occasionally cause the native VMTK sectioning step to fail, typically as an internal kernel error. To guarantee a complete, continuous area profile, a robust fallback replicates the same orthogonal slicing logic with Visualization Toolkit (VTK) operators when a failure is detected. At each evaluation point along the centerline, a `vtkPlane` (oriented perpendicularly to the centerline) combined with a `vtkCutter` slices the 3D surface to extract the cross-sectional lumen contour and compute its enclosed area in mm^2 using a standard polygon formula. This ensures accurate cross-sectional areas even in the most challenging anatomical segments.

Phase 2: Tabular Area Integration and Branch Mapping

To perform localized stenosis quantification, the cross-sectional areas computed across the global coronary tree must be accurately mapped to the individual ostium-to-endpoint branch dataframes generated in Block 1.

For every spatial coordinate (P_x, P_y, P_z) in a branch table, if the sequence of centerline points aligns perfectly with the global area output, the value is assigned via direct row matching. If the points do not align exactly, the system uses a `scipy.spatial.cKDTree` nearest-neighbour query to find the closest evaluated station in 3D space and map its computed area to the specific branch point.

Ultimately, this mapping step propagates a continuous `Area` metric to every discrete coordinate across all branch paths. It seamlessly merges the newly extracted geometry with the anatomical information inherited from Block 1, preparing the exact dataset required for the next baseline reference calculations.

Phase 3: Geodesic Distance and Reference Area Estimation

Stenosis severity is evaluated independently along each individual branch path rather than on the global coronary tree dataset. This isolation is critical: it ensures that spatial evaluation strictly follows the natural, antegrade blood flow from the ostium to the distal tip. Evaluating the tree globally could inadvertently select reference points that are spatially close in Euclidean space but belong to entirely different, non-connected vessels.

To measure distance along the tortuous 3D vessel, consecutive centerline points are connected by straight segments, and the cumulative sum of these segment lengths defines a one-dimensional geodesic distance (g_d). This g_d value is then integrated into the branch dataframe for every discrete coordinate.

To evaluate stenosis at any specific point, the algorithm requires a healthy reference area for comparison. The system first identifies proximal and distal centerline points located at geodesic positions $g_d - W$ and $g_d + W$, respectively. The reference half-window W is a crucial hyperparameter in the pipeline. Based on the geometric stenosis framework introduced by Burrull [7] and standard angiographic principles described by Hideo-Kajita et al. [16], W is set to 10 mm in this project. However, this 10 mm threshold is an empirical approximation: there is no universally optimal value, as the ideal reference distance depends on patient-specific anatomy and individual lesion length. Ideally, to maximize clinical accuracy, the system would allow a physician to manually select or adjust proximal and distal reference points based on visual anatomical assessment. Because the computed stenosis varies significantly with the automated window size, this introduces an important methodological limitation that is discussed later.

Once the two reference points are identified along the same branch, their corresponding mapped areas (A_{prox} and A_{dist}) are retrieved. The healthy reference area (A_{ref}) at the evaluation point is then calculated as the arithmetic mean of these bilateral values:

$$A_{\text{ref}} = \frac{A_{\text{prox}} + A_{\text{dist}}}{2}. \tag{2.2}$$

Because this formula requires both proximal and distal context, percentage area stenosis cannot be calculated for points at the very extremities of branch paths. Consequently, any evaluation point lying within distance W from the ostium or the distal tip is excluded, as it would possess only a one-sided reference. Alongside sensitivity to window size, additional anatomical challenges of this automated baseline (such as unpredictable behaviour near major bifurcations) are addressed in Chapter 4.

Phase 4: Percentage Area Stenosis and Patient-Level Aggregation

For each qualifying centerline point, the pipeline quantifies stenosis severity. The percentage area stenosis (%AS) formula, adapted from Burrull [7], compares the local cross-sectional lumen area (A) directly against its corresponding healthy reference (A_{ref}):

$$\%\text{AS} = \left(1 - \frac{A}{A_{\text{ref}}}\right) \times 100. \tag{2.3}$$

Next, the separate branch tables are merged into a unified patient record. Since all extracted branches trace continuously from the ostium to a distal endpoint, they inherently share identical coordinates along common arterial trunks. When these tables are combined, spatial deduplication is applied based on rounded (P_x, P_y, P_z) tuples. If multiple overlapping points exist at the same coordinate, the algorithm strictly retains the row with the highest %AS value. This conservative rule ensures that no maximum stenosis is hidden, preventing clinical underestimation of lesion severity where anatomical paths overlap.

Ultimately, Block 2 yields two structured outputs: the enriched individual branch datasets (containing g_d, A_{prox}, A_{dist}, A_{ref}, and %AS, among others) and the deduplicated global coronary tree dataset. These comprehensive tables constitute the definitive quantitative input consumed by Block 3.

2.5. Block 3: Artery Labeling and CAD-RADS 2.0 Scoring

Block 3 translates the quantitative stenosis metrics produced in Block 2 into a standardized clinical classification using the Coronary Artery Disease Reporting and Data System (CAD-RADS 2.0) framework [12].

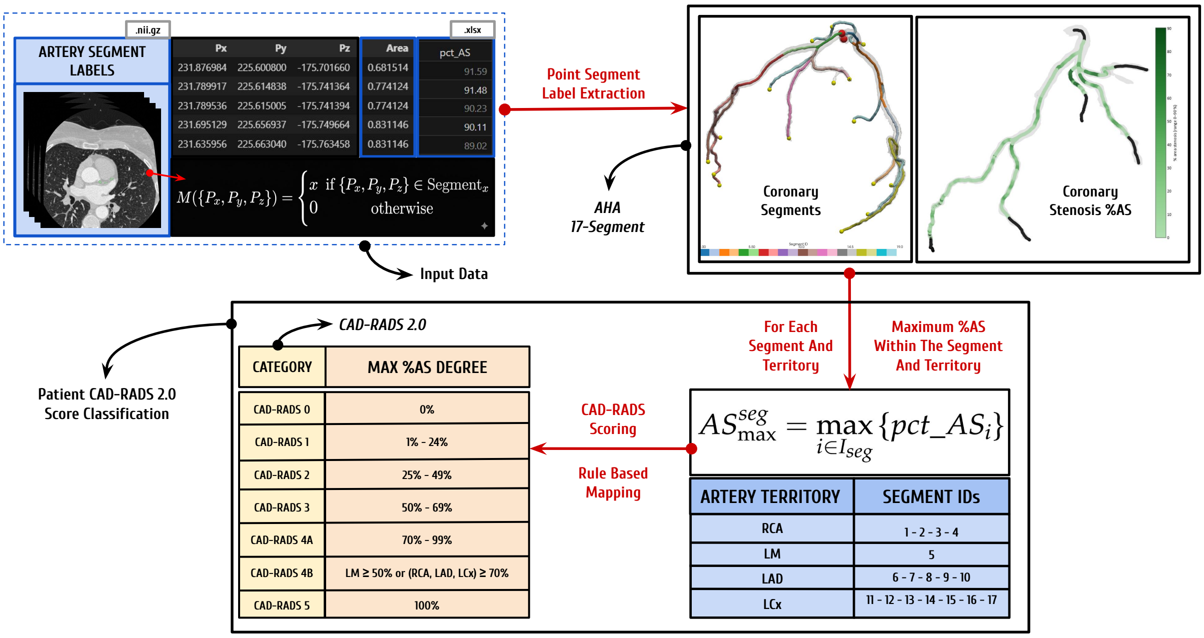

Figure 2.4: Block 3 workflow: segment aggregation based on the AHA 17-segment model and automated CAD-RADS 2.0 scoring.

As illustrated in Figure 2.4, this block ingests the merged patient-level stenosis table from Block 2 together with the anatomical segment labels inherited from Block 1. The pipeline first summarizes the stenosis severity at the individual segment level. These segments are subsequently grouped into the primary coronary territories (RCA, LM, LAD, and LCx), establishing the essential anatomical framework required for the CAD-RADS 2.0 evaluation. Finally, the system applies rule-based logic to derive the overall patient-level CAD-RADS category and calculates the Segment Involvement Score (SIS), structuring these findings into the definitive outputs consumed by Block 4.

2.5.1. Anatomical Context: AHA 17-Segment Model

Reliable CAD-RADS assignment requires mapping each evaluated centerline point to a recognized coronary territory. The pipeline adopts the 17-segment American Heart Association (AHA) model, a widely used reporting standard for regional coronary anatomy [5]. This choice is not only clinically established but also technically aligned with the voxel labeling scheme of the primary cohorts used in this thesis: integer labels in the ASOCA annotation volumes [15] are consumed directly as segment ID values without intermediate remapping. This direct compatibility preserves pipeline robustness and keeps segment-level aggregation consistent across patients.

Beyond individual anatomical segments, this model maps every branch into one of four primary coronary territories: the Right Coronary Artery (RCA), Left Main (LM), Left Anterior Descending (LAD), and Left Circumflex (LCX). This territorial grouping is a fundamental requirement of the CAD-RADS 2.0 scoring framework, which assigns the patient-level category from the most severe stenosis observed across these vessels [12]. Table 2.2 summarizes the complete segment-to-territory dictionary used throughout Block 3 and the clinical dashboard. Background voxels are encoded as segment 0 and are subsequently excluded from stenosis aggregation.

Table 2.2: AHA 17-segment dictionary used for anatomical labeling and CAD-RADS territory grouping. Adapted from Austen et al. [5].

ID	Abbreviation	Anatomical name	Territory
1	pRCA	Proximal RCA	RCA
2	mRCA	Mid RCA	RCA
3	dRCA	Distal RCA	RCA
4	rPDA	Right PDA	RCA
5	LM	Left Main	LM
6	pLAD	Proximal LAD	LAD
7	mLAD	Mid LAD	LAD
8	dLAD	Distal LAD	LAD
9	D1	First diagonal	LAD
10	D2	Second diagonal	LAD
11	pLCX	Proximal LCX	LCX
12	OM1	First obtuse marginal	LCX
13	dLCX	Distal LCX	LCX
14	OM2	Second obtuse marginal	LCX
15	lPDA	Left coronary PDA	LCX
16	L-ILB	LCX inferolateral branch	LCX
17	L-PLB	LCX posterolateral branch	LCX

2.5.2. Segment-Level Stenosis Aggregation

For each patient, all centerline rows in the Block 2 merged table that carry a valid `Segment_ID` are grouped by segment identifier using tabular aggregation. Within each segment, the algorithm retains the absolute maximum %AS, ensuring that focal lesions are not underestimated when multiple samples exist along the same anatomical region. Each segment is then assigned a discrete stenosis severity grade aligned with the CAD-RADS luminal stenosis bins (see Table 2.3).

Any `Segment_ID` present in the input data but outside the 1–17 dictionary (for example, labels introduced by extended annotation schemes) is flagged as *unmapped*. These rows may still appear in exploratory outputs, but they are excluded from the standard territory groupings used for CAD-RADS 2.0 triage, preventing execution failure while making non-standard labels explicit.

2.5.3. Patient-Level CAD-RADS 2.0 Classification

Patient-level scoring follows the CAD-RADS 2.0 decision principles [12]. The algorithm first determines the highest segment-level stenosis grade from the aggregated table. This grade is derived from the maximum %AS observed across all evaluated segments and subsequently classified into the standard luminal categories detailed in Table 2.3. Total coronary occlusion (category 5) is assigned when any segment reaches 100% %AS.

Categories 4A and 4B distinguish high-risk patterns among severe stenoses. As established by the CAD-RADS 2.0 framework [12], category 4B is assigned when either (i) left-main (segment 5) stenosis is at least 50% %AS, or (ii) obstructive disease ($\geq 70\%$ %AS) is present in each of the three major territories (RCA, LAD, and LCX) defined in Table 2.4. If severe stenosis is present but neither of these criteria is met, the patient is classified as CAD-RADS 4A. Categories 0–3 are assigned directly from the highest segment grade when no category 4 or 5 rule overrides the result.

Table 2.3: CAD-RADS 2.0 luminal stenosis categories applied to maximum segment-level %AS. Adapted from Cury et al. [12].

Category	Max. %AS (segment)	Clinical interpretation (summary)
0	0%	No visible stenosis
1	1–24%	Minimal stenosis
2	25–49%	Mild stenosis
3	50–69%	Moderate stenosis
4A	70–99%	Severe stenosis without CAD-RADS 4B criteria
4B	70–99% (territory pattern) or LM $\geq 50\%$	High-risk severe disease pattern
5	100%	Total coronary occlusion

Table 2.4: Major coronary territories used for CAD-RADS 2.0 triage rules in this pipeline (segment IDs follow the AHA 17-segment model). Adapted from Austen et al. and Cury et al. [5, 12].

Territory	Segment IDs	Role in automated scoring
RCA	1, 2, 3, 4	Obstructive/severe flags per vessel; contributes to three-vessel 4B rule
LM	5	Left-main $\geq 50\%$ %AS triggers CAD-RADS 4B
LAD	6, 7, 8, 9, 10	Obstructive/severe flags per vessel; contributes to three-vessel 4B rule
LCX	11, 12, 13, 14, 15, 16, 17	Obstructive/severe flags per vessel; contributes to three-vessel 4B rule

Finally, the system calculates the Segment Involvement Score (SIS), defined as the absolute number of assessable coronary segments exhibiting any measurable atherosclerotic plaque [12]. Block 3 concludes by exporting a comprehensive set of structured outputs: a segment-level stenosis summary, a patient-level CAD-RADS report (detailing the assigned category, classification rationale, highest lesion location, territory flags, and SIS), and the geometrically enriched branch tables. All these generated metrics and records are specifically formatted to serve as the direct quantitative input for the interactive clinical visualization dashboard detailed in Block 4.

2.6. Block 4: Patient Visualization Dashboard

2.7. Experimental Design and Validation Framework

2.7.1. Workflow Validation on Real Datasets: ASOCA

2.7.2. Workflow Validation on Real Datasets: MACS-18

2.7.3. Geometry and Quantification Verification: Synthetic Data

2.7.4. User Experience and Interface Usability Evaluation

Chapter 3

RESULTS

Chapter 4

DISCUSSION

Chapter 5

CONCLUSIONS

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Appendix A

SUPPORTING MATERIAL